Lorenzo Corti

Residence: Netherlands – **Nationality**: Italian – Authorised to work in the EU.

Website: https://lcorti.github.io/ — **Email**: corti.lorenz@gmail.com — **LinkedIn**: https://www.linkedin.com/in/lorenzo-corti-46a22b181/
GitHub: https://github.com/LCorti — **Scholar**: https://scholar.google.com/citations?user=4NHtXOUAAAAJ

Summary

Applied AI engineer (PhD Feb. 2027), specialised in NLP and in evaluating LLM and vision-language model (VLM) systems. Developed DiaVLo (EMNLP 2026), a causal framework to attribute VLM outputs to visual concepts, and tested on 4 VLMs across 4 benchmarks. Built an LLM behaviour-testing toolkit validated with NLP industry practitioners, and delivered NLP pipelines and platforms (IDiOM, PERSEUS, COCTEAU) covering data ingestion, transformer-based analysis, and web deployment. 4+ years of hands-on experience with PyTorch, Transformers, spaCy, and ML & DL frameworks.

Skills

- **LLM & VLM**: Transformer architectures (encoder, encoder-decoder, decoder-only), prompt engineering, visual question-answering (VQA), named entity recognition (NER), topic modelling, text classification, text generation.
- **ML & DL Frameworks**: PyTorch, Hugging Face (Transformers, Datasets, Evaluate), spaCy, scikit-learn, Captum, NumPy, Pandas, SentenceTransformers, OpenCV, TensorFlow.
- **AI Evaluation**: LLM and VLM benchmarking, model behaviour testing, explainable AI, explanation fidelity evaluation.
- **Quantitative Methods**: Causal inference (DoubleML, DoWhy, EconML), causal discovery (CDT, NetworkX).
- **Back-end & Data**: FastAPI, Django, Spring Boot, MongoDB, MySQL, Elasticsearch.
- **Infrastructure**: AI model self-hosting, Docker, Git, GCP, VPS.
- **Programming Languages**: Python, Java, C, Javascript.
- **Technical Leadership**: Mentored 8 M.Sc. theses, 5 B.Sc. theses, and 10 student group projects on NLP, language models, interpretability, human computation, and information systems.

Experience

Doctoral Researcher, Delft University of Technology **Feb 2022 – Feb 2027**

Diagnosis of Vision-Language Model Behaviours

- Developed DiaVLo, a framework to verify observable VLM behaviours against specifications of desired behaviour and to quantify concept-level contributions to VLM outputs. Contributions are estimated with a Double-ML causal estimator fitted to synthetically occluded visual inputs. Applied it to 4 VLMs across 4 benchmarks for multiple-choice and open-ended visual question answering.
- Self-hosted VLMs with Hugging Face Transformers and PyTorch; scene-graph extraction with IETrans and NetworkX; causal analysis with DoWhy and EconML; processing with OpenCV, SentenceTransformers, spaCy, NumPy and scikit-learn.

Toolkit for LLM Behaviour Diagnosis

- Designed and built a toolkit that diagnoses LLM behaviours on known linguistic structures. It supports encoder-only, encoder-decoder and decoder-only transformer LLMs, multiple NLP datasets and standard NLP metrics, with inspection and tagging of LLM outputs, behavioural testing and reporting.
- Implemented system: FastAPI + MongoDB backend, Django + MySQL frontend, Docker containerisation for self-hosting; built on Hugging Face Transformers/Datasets, PyTorch, spaCy, Captum, NumPy, Pandas, and scikit-learn.
- Validated externally with 14 NLP industry practitioners.

Extraction of Behaviours of Vision Models

- Co-developed CHIME, an explainable AI method to extract observable behaviours of CNN-based vision models. Applied it to models with known biases injected via fine-tuning; evaluated fidelity, coherence, and accuracy of the behaviours.
- Causal estimation with DoWhy; causal graph discovery and analysis with CDT and NetworkX; analysis with NumPy, Pandas, scikit-learn and Matplotlib.

Research Assistant, Politecnico di Milano **Feb 2020 – Jan 2022**

IDiOM: NLP pipeline for disinformation analysis

- Built a modular and open-source NLP data analysis pipeline for disinformation. Implements data collection from social and news media sources and runs NLP and ML analyses. Stack: Hugging Face Transformers, spaCy, scikit-learn, NumPy, MongoDB, custom ingestion/cleaning/processing modules.

PERSEUS: document search and analysis platform

– Built a platform using language models for NER, topic extraction and document analysis. Stack: Python FastAPI, Hugging Face Transformers, scikit-learn, Elasticsearch, jQuery/Bootstrap frontend, third-party APIs.

GANs for Time Series Imputation

– Training from scratch of GANs combining LSTM and convolutional networks; implemented multiple training regimes and normalisations to prevent mode collapse (TensorFlow, NumPy, scikit-learn).

VaccinItaly: analysis of Italian COVID-19 discourse

– Analysed vaccine perception and hesitancy on Twitter, Facebook and news outlets (temporal, geographical, lexical analyses) and trained pro-/anti-vaccination stance classifiers, using the IDiOM pipeline, scikit-learn, and spaCy.

COCTEAU: gamified crowdsourcing platform

– Developed a platform for large-scale citizen engagement. Stack: Java, Spring Boot, jQuery, Bootstrap, MySQL, third-party APIs, Google Cloud Platform.

Education

– PhD Candidate in Computer Science, Delft University of Technology, Feb 2022 – Feb 2027
– M.Sc. in Computer Science and Engineering, Politecnico di Milano, Sept 2017 – Dec 2019
– B.Sc. in Computer Science and Engineering, Politecnico di Milano, Sept 2014 – July 2017

Languages

– **Italian**: Native.
– **English**: Full Professional Proficiency.
– **Spanish**: Advanced.
– **Dutch**: A1 (learning).

Selected Publications

1. *DiaVLo: Diagnosing Behaviours of Vision-Language Models*. **Lorenzo Corti**, Jie Yang. Findings of the Association for Computational Linguistics: EMNLP 2026. Link: https://arxiv.org/abs/2609.22008.
2. *Toolkit for LLM Behaviour Diagnosis*. **Lorenzo Corti**, Sebastiaan Beekman, Jie Yang. *Under review*.
3. *A.I. Robustness: a Human-Centered Perspective on Technological Challenges and Opportunities*. Andrea Tocchetti*, **Lorenzo Corti***, Agathe Balayn*, Mireia Yurrita, Philip Lippmann, Marco Brambilla, Jie Yang (* *equal contribution*). In: ACM Computing Surveys. 2025. DOI: https://doi.org/10.1145/3665926.
4. *ARTIST: ARTificial Intelligence for Simplified Text*. **Lorenzo Corti**, Jie Yang. In: Generative AI and HCI workshop at CHI. 2023. DOI: https://doi.org/10.48550/arXiv.2308.13458.
5. *CHIME: Causal Human-in-the-Loop Model Explanations*. Shreyan Biswas, **Lorenzo Corti**, Stefan Buijsman, Jie Yang. In: Proceedings of the AAAI Conference on Human Computation and Crowdsourcing. 2022. DOI: https://doi.org/10.1609/hcomp.v10i1.21985.